

Bugs found: 4

1. Double free error: `free(p)` is called twice on the same memory address.
2. Use-after-free error: the pointer '`p`' is dereferenced in `printf` after being freed.
3. Uninitialized variable: the integer '`x`' is used in `printf` without being assigned a value.
4. Missing check for malloc failure: the return value of `malloc` is not verified to be non-NULL before dereferencing.